2. USER PERSONAS

2.1 What Is a Persona?

Simple Definition: A persona is a **made-up character** that represents a real group of your users. It's not one real person—it's a mix of many real people you've studied.

Think of it like this: Instead of saying "our users are students aged 18-25 who use phones", you create "Ali - a 20-year-old university student who orders food while studying".

Why do we need personas?

- You can't design for "everyone" (designing for everyone = designing for no one)
- Personas help you focus on **specific real people** instead of imaginary general users
- It's easier to remember 4 characters than 4000 survey responses
- When making decisions, you can ask "Would Ali like this feature?"

Remember This:

- Persona = Fake person based on real research
- Purpose = Design for a specific somebody, not generic everybody

2.2 How to Create Personas: 8 Steps (Made Simple)

Step	What To Do	Example
1. Decide sample size	Figure out how many people to interview/survey	"We'll survey 50 students"
2. Collect data	Use surveys, interviews, watching users	Ask users about their problems
3. Find patterns	Read all responses, note what's similar	"Most users complain about slow loading"
4. Create 3-6 personas	Group similar users into character profiles	Budget students, Working professionals
5. Validate	Interview real users to check if personas are accurate	"Does this description match you?"
6. Categorize	Put each studied user into one persona	"This user is like Budget Student Bilal"
7. Note outliers	Track users who don't fit any persona	Helps know who you're NOT designing for
8. Name them	Give memorable names	"Budget Bilal", "Professional Priya"

Key Numbers: Keep **3 to 6 personas** (less than 3 = too simple, more than 6 = hard to remember)

2.3 What Goes in a Persona Template?

Part	What It Means	Example
Name & Photo	Made-up name + stock photo	"Budget Bilal" + picture
Demographics	Age, job, location (only relevant ones)	22, Student, Karachi
Goals	What they want to achieve	"Order cheap food fast"
Frustrations	What annoys them	"Apps are too slow"
Behaviors	How they currently do things	"Always checks prices first"
Tech Comfort	Good with technology or not?	"Uses phone 6hrs/day"
Quote	A sentence that captures their thinking	"I don't have time to waste"

5.8 HEURISTIC EVALUATION

What Is It?

Simple Definition: Experts check your app against a list of usability rules (called "heuristics") to find problems.

Think of it like: A checklist inspection for your app's usability.

Nielsen's 10 Rules (Easy Version)

#	Rule	What It Really Means	Real Example
1	**Show what's happening**	Tell users what the system is doing	WhatsApp blue ticks show "message seen"
2	**Use user's language**	Don't use technical words	Say "Save" not "Commit to database"
3	**Give users an exit**	Let users undo mistakes easily	"Cancel" button, Ctrl+Z
4	**Be consistent**	Same action = same result everywhere	"Delete" should always delete, not "Remove" sometimes
5	**Prevent errors**	Stop mistakes before they happen	Grey out "Submit" until form is complete

6	**Show options, don't make users memorize**	Display choices instead of expecting users to remember	Dropdown menu vs typing from memory
7	**Shortcuts for experts**	Fast methods for experienced users	Ctrl+C to copy instead of menu
8	**Keep it clean**	Remove unnecessary stuff	Don't show rarely-used options on main screen
9	**Help with errors**	If error happens, explain clearly	"Wrong password" not "Error 401"
10	**Provide help**	Add documentation if needed	Help page, tooltips, FAQs

6. ETHNOGRAPHIC TECHNIQUES (OBSERVATION METHODS)

6.1 What Is Observation?

Simple Definition: Watching users to see what they actually do (not what they say they do).

Why watch users?

- People often do things differently than they describe
- You see the real environment, real problems, real frustrations
- You notice things users forgot to mention

What you learn from watching:

- What users actually do (real behavior)
- Where they do it (environment)
- Where the app helps or fails them
- What extra help they need

6.2 Where Can You Observe?

Option 1: Lab Testing (Controlled)

- Special room with cameras, one-way mirror
- You give users specific tasks to do
- You measure time, errors, clicks
- **Pros:** Easy to compare results, get exact numbers
- **Cons:** Fake environment, not real-life behavior

Option 2: Field Study (Natural Environment)

- Watch users where they actually work/live
- Real tasks, real interruptions, real context
- **Pros:** See true behavior, discover unexpected things
- **Cons:** Hard to control, hard to compare

Which is better? Neither! Use lab for numbers, field for understanding. Best approach: use both.

6.3 Outsider vs Insider: How Close Do You Get?

When observing users, you can be:

Position	What You Do	Pros	Cons
Pure Outsider	Watch from outside, don't participate	Completely objective	Miss context, don't understand why
Observer + Participant	Watch while occasionally joining	Balance of both	Can be awkward
Full Insider	Become part of the group	Deep understanding	May lose objectivity

Real-World Examples:

- **Outsider:** Security camera watching shoppers
- **Participant Observer:** New employee learning the job
- **Full Insider:** Working undercover in a company for months

6.4 Types of Observation (Quick Summary)

Type	What It Is	When To Use
Quick & Dirty	Fast, informal watching and chatting	Need quick feedback, limited time
Usability Testing	Formal lab with recordings	Need exact measurements
Field Study	Watch in real environment	Need to understand real context
Participant Observation	Join the users, do what they do	Need deep understanding
Ethnography	Long-term deep study (weeks/months)	Academic research, major projects

6.5 Lab vs Field: Side-by-Side Comparison

Example: Testing a Phone App for Finding Restaurants

Aspect	Field Study	Lab Study
Setup	Join real people looking for a real restaurant	Give task: "Find phone number of Restaurant X"
Task	Real need (actually hungry!)	Artificial task
Results	See frustration, button size issues, screen problems	Get exact time measurements, error counts
Pros	Real situation, real feelings	Can compare users, calculate averages
Cons	Not objective, can't compare easily	Misses real-world context

Bottom Line: Field shows you the "why", Lab gives you the "how much"

7. APPLIED PSYCHOLOGY OF THE USER

7.1 The User Behavior Formula

Why users do what they do = 4 things:

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USER BEHAVIOR = Goals + Task Structure + Knowledge + Brain Limits
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Let me explain each:

Component	What It Means	Example
Goals	What user wants to achieve	"I want to order pizza"
Task Structure	Steps the app makes them do	Register → Login → Browse → Add to cart → Pay
Knowledge	What user already knows	"I know how online shopping works"
Brain Limits	Memory limits, tendency to make mistakes	Can only remember 7 things at once

7.2 How to Improve User Experience

Based on the formula, you can improve things by changing each part:

Strategy	How To Do It	Example
Improve Task Structure	Make steps easier, reduce steps	One-click checkout instead of 5 forms
Increase Knowledge	Teach users, give tutorials	Show tooltips for new features
Control Errors	Add undo, confirmations, prevent mistakes	"Are you sure you want to delete?"
Organize Goals	Break big tasks into small ones	Progress bar showing Step 1 of 5

Important Brain Facts:

- Humans can only remember **7±2 items** at once (Miller's Law)
- People make random mistakes even on tasks they know well
- **Solution:** Break big tasks into small steps → less to remember, smaller mistakes

7.3 How to Measure User Behavior

Dimension	What It Measures	Example Question
Functionality	What can the system do?	"Can it do what I need?"
Learning	How long to learn?	"How many hours to become good?"
Time	How fast to complete tasks?	"How many seconds to send a message?"
Errors	How many mistakes?	"How often do users click wrong button?"
Quality	How good are results?	"Does it produce what user wanted?"
Robustness	Can handle new situations?	"Can user figure out new features?"
Acceptability	Do users like it?	"Would they recommend it?"

Trade-offs: These often conflict. Faster might mean harder to learn. More features mean more complex.

7.4 Novice vs Expert Users

Aspect	Novice (New User)	Expert (Experienced User)
Main concern	"How do I do this?"	"How do I do this FAST?"
Needs	Step-by-step guidance, tutorials	Shortcuts, customization
Annoyed by	Too many options, confusing menus	Unnecessary confirmations, forced tutorials
Design for them	Simple defaults, limited choices	Hidden advanced features, keyboard shortcuts

7.5 Two Views of User Interface

Designer thinks: UI = screen + buttons + code

Psychology says: UI = **everything user touches, sees, or thinks about**

This includes:

- The screen (obvious)
- Documentation and help (often forgotten!)
- User's mental picture of how it works
- Training materials

Key Point: UI design must start early, not be added at the end!

7.6 Conceptual Models

What is it? The mental picture users have of how your system works.

Why it matters:

- If their mental picture matches reality → app feels easy
- If their mental picture is wrong → confusion and errors

Example:

- User thinks "trash" permanently deletes immediately
- But system has 30-day recovery
- Mismatch = user confusion

Good design = make the mental model obvious and consistent

8. TECHNOLOGY ACCEPTANCE MODEL (TAM)

8.1 What Is TAM?

TAM explains **why people accept or reject technology**.

The Big Discovery: *"Usefulness" is 50% MORE important than "Ease of Use"*

What this means:

- People will struggle with a difficult app IF it helps their work
- People will abandon an easy app IF it doesn't help their work
- Best: Be BOTH useful AND easy

8.2 Two Key Terms

Term	Definition (Simple)	Example
Perceived Usefulness	"Does this help me do my job better?"	"This app saves me 2 hours daily"
Perceived Ease of Use	"Is this hard to use?"	"I figured it out in 5 minutes"

8.3 How TAM Works (Flow)

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Design Features → Ease of Use → Usefulness → Attitude → Actual Use
                                ↓               ↑
                                └───────────────┘
                              (Easy = feels MORE useful)
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Key Points:

1. Easy to use → feels MORE useful (but not the other way)
2. Design features don't directly cause use—they work through usefulness/ease
3. Both usefulness AND ease affect attitude → attitude affects actual use

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8.4 What This Means for Designers

To make a system more acceptable:

Option	How	When to use
Add useful features	Give users new capabilities	When users need to do more
Make existing features easier	Reduce steps, improve flow	When users struggle with current features

The Golden Rule: *Users will tolerate a hard interface that helps them work. Users will reject an easy interface that doesn't help them work.*

9. UNIVERSAL USABILITY

9.1 What Is Universal Usability?

Goal: Make computers usable by 90% of all households.

Current Problem:

- People waste **5.1 hours per week** fighting with computers
- That's more time wasted than on traffic jams!
- Many people give up and don't use computers at all

9.2 The Three Big Challenges

Challenge	What It Means	Examples
1. Technology Variety	Different devices, screens, speeds	Phone vs laptop, fast vs slow internet
2. User Diversity	Different people with different abilities	Kids, elderly, blind, experts, beginners
3. Knowledge Gaps	Different skill levels	Some learn in 5 minutes, others need 5 hours

9.3 E-Government Example

Government websites must work for EVERYONE because:

- Can't exclude citizens based on tech skill
- Everyone has different backgrounds
- High stakes (taxes, IDs, services)

9.4 Counter-Argument to Critics

Critics say: "Designing for disabled/unskilled users makes the app worse for everyone else"

Response: WRONG! Designing for more people creates better solutions for ALL.

Proof (Real Examples):

Designed For	Also Helps
Wheelchair ramps (curb cuts)	Parents with strollers, luggage carts, bicycles
Subtitles (for deaf users)	People in noisy places, language learners
Voice control (for blind users)	Drivers, people cooking with dirty hands

9.5 Strategies for Universal Usability

1. Layered Approaches (Give Options)

- Beginner mode vs Advanced mode
- University website: Separate sections for students, faculty, alumni

2. Design for Different Users

User Type	What They Need
Poor readers	Simple text, pictures, multiple languages
Elderly/poor vision	Big fonts, high contrast
Blind users	Screen reader support, keyboard navigation
Expert users	Shortcuts, customization, macros

3. Bridge Knowledge Gaps

User Type	Solution
Novice	Step-by-step guides, simple tutorials, easy undo
Sometimes-users	Context-sensitive help, reminders, consistent design

Expert	Shortcuts, dense information, minimal interruptions
Everyone	Online help, phone support, community forums